# GEOMETRIC REDUCEDNESS SATISFIES THE FIVE PROPERTIES REQUIRED OF A FORMAL FIBRE CONDITION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Editorial note.* The author's proof is a checklist: one line per property, each naming the exact result it follows from (definitions or named lemmas for the local characterisation, ascent, descent and separable algebraic base change). Terse but complete, not a hint-only proof.

*Allowed standard prerequisites (editorial).*

- Definition of geometric reducedness. Algebra, Definition \ref{algebra-definition-geometrically-reduced} and Algebra, Lemma \ref{algebra-lemma-reduced-goes-up}, cited at proof lines 13204 and 13208. Give properties (A) and (C).
- Reducedness descends along ring maps, and geometric reducedness is preserved by separable algebraic base change. Algebra, Lemma \ref{algebra-lemma-descent-reduced} and Algebra, Lemma \ref{algebra-lemma-geometrically-reduced-over-separable-algebraic}, cited at proof lines 13210-13212. Give properties (D) and (E).

*Source reference.* Original lemma 0BIW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14504–14504. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 030S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3959–3959. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 07QK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9835–9835. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 033F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4054–4054. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0C2Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15173–15173. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Properties (A), (B), (C), (D), and (E) hold for $P(k \to R) =$"$R$ is geometrically reduced over $k$".*

*Proof.* Part (A) follows from the definition of geometrically reduced algebras (Algebra, Definition 030S). Part (B) follows too: a ring is reduced if and only if all local rings are reduced. Part (C). This follows from Lemma 07QK. Part (D). This follows from Algebra, Lemma 033F. Part (E). This follows from Algebra, Lemma 0C2Y. □